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# Becoming a conversationalist: Questions, challenges, and new directions in the study of child interactional development

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Interaction with others is a primary basis for linguistic and social development (e.g., Beebe et al., 1997; Bruner, 1983; Leong & Schilbach, 2019). As such, it is critical to understand how and when children acquire the skills for coordinated interaction and to examine how interactional development influences (and is influenced by) other aspects of linguistic, social, and cognitive development.

The course of interactional development is not yet well understood. Children begin to demonstrate basic interactional skills in early infancy, but years pass before their conversational contributions approximate that of adults' (e.g., Casillas et al., 2016; Dunn & Shatz, 1989; Ervin-Tripp, 1979; Hilbrink et al., 2015; Jaffe et al., 2001; Stivers et al., 2018). How and why this arc of development is so prolonged has yet to be explained. Meanwhile, we have seen that US children's experience with contingent language predicts variation in their linguistic and neurocognitive development, suggesting a tight link between children's use of their turn-taking skills and other areas of development (e.g., Donnelly & Kidd, 2021; Ferjan Ramírez et al., 2020; Gilkerson et al., 2018; Romeo et al., 2018, 2021). However, the mechanisms by which this relationship arises and the generalizability of these findings are still unknown.

Prior findings on interactional development raise a number of important questions. For example, with respect to the predictive value of early turn taking for later linguistic and neurocognitive development, we have yet to learn how these findings apply in communities where typically developing children are brought up with infrequent child-directed talk from adults, but frequent peer talk or overheard talk. Another important question is how digital communication platforms play a role in children's interactional development (e.g., video chat; (Roseberry et al., 2014; Roche et al., 2022). We might ask how these interactional experiences influence children's development, and how children develop platform-specific skills and expectations for conducting interaction (e.g., initiating and maintaining joint attention). Big data approaches to measuring children's naturalistic language environments holds great promise for establishing critical constraints on theories of children's language learning (e.g., via automated analysis of continuous audio/video recording of whole waking days; (Ferjan Ramírez et al., 2021; Greenwood et al., 2011; Wang et al., 2020). However, to achieve analysis at scale, current automated approaches sacrifice important information about how children achieve interaction with others (e.g., by not capturing visual signals or by classifying all nearby speech as potentially contingent)—information that is readily available with manual annotation but thus limited in scale.

The articles included in this special issue touch upon these issues and more, investigating interaction in the domains of contingent caregiver response, multimodal infant interaction, social attention, joint attention, and predicting linguistic and cognitive outcomes in

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the first three years of life. Collectively, these papers represent 983 participants from multiple linguistic and cultural backgrounds in the U.S., Turkey, Poland, Canada, the U.K., and the Gambia. While naturalistic interaction was at the heart of this collection of studies, the papers feature a diverse array of methods for studying infants' social interactions (in-lab/at-home; brief/daylong; in person/remote; manual/automated annotation). Developmental outcomes in these studies focused on both infant vocal learning and later vocabulary development, with nearly every study showcasing longitudinal analyses, thus shedding light on how early interactions influence later language development.

Each study also raises specific open questions, challenges, and new directions in the study of child conversational and interactional development. Among the submissions, we identified four major themes: how early coordinated interaction predicts later language outcomes, multimodal features of early communication, coordinating and sharing attention during interaction with others, and variation in interactional development.

## 1. Coordinated interaction and language development

Four papers examined how early coordinated interaction predicts children's later language outcomes. Zhang et al. (2024) analyzed free-play sessions of $N = 30$ U.S. English speaking parents and their infants. They report that prelinguistic vocal turn taking is positively related to early language development. Specifically, caregiver responsiveness to babbling at 5 months facilitated vocal turn taking at 10 months; and caregiver responsiveness at 10 months predicted parent-reported expressive vocabulary at 18 months. In another study, Chen et al. (2023) examined associations between "serve and return" parent-child interactions and children's language skills at 18 and 24 months in a diverse population (mainly Hispanic) of $N = 148$ U.S. families. They noted that both maternal meaningful responses and paternal prompt responses to their child's vocalizations at 9 months were positively associated with expressive language at 18 months, while interactions were observed between maternal and paternal effects on child language at 24 months. This study uniquely highlights the inter-related contributions that mothers and fathers make to their child's language development.

Honing in on the particular contribution of maternal infant directed speech, Endevelt-Shapira et al. (2024) use longitudinal data (3–30 months) from $N = 40$ U.S. English-learning participants in order to draw links between infants' interactional development and their later vocabulary size. For each infant, maternal sensitivity (in-lab) and interactive language input (at-home) were evaluated via manual annotation of natural interaction at 3 months; parent-reported expressive vocabulary size was then collected regularly between ages 18 and 30 months. While a number of early interactional behaviors were associated with later vocabulary size, the results point toward particularly clear effects of maternal infant directed speech in one-on-one interactions on children's later vocabulary. The authors argue for the special role that these interactions play in shaping infants' social attention to language during the early phase of language development.

Taking a closer look at family structure and its effects on the interactional environment, Katus et al. (2024) obtained naturalistic seven-hour-long LENA recordings at 12, 18, and 24 months of age from a cohort of $N = 204$ infants from Mandinka-speaking households in The Gambia and $N = 61$ infants from English-speaking households in the UK. Following a validation of automated LENA outputs in the Gambian cohort, the authors showed that contingent turn taking at 12 and 18 months predicted later child vocalizations at 18 and 24 months. Interestingly, the number of caregivers showed an inverse U-shaped relationship with contingent turn-taking, with highest turn-taking observed for an intermediate number of caregivers. This study provides valuable insights into the context of child caregiving in The Gambia and its associations with child language development.

## 2. Multimodal features of early communication

How do multimodal signals support early interactions? Using in-home recordings, Karadağ et al. (2024) detail a wide range of socio-communicative behaviors of $N = 43$ Turkish 18-month old infants. Across a wide range of SES backgrounds, these infants were found to actively communicate with their social partners to fulfill their need-based and non-need-based motivations using a wide range of verbal and nonverbal behaviors. In another study, Beebe et al. (2024) used detailed coding of interactive behavior in $N = 100$ U.S. Latina mother-infant pairs to reveal predictive associations between the temporal dynamics of their contingent coordination of gaze, affect, and touch at four months, and infant cognitive performance at one year. The findings showed predictive relationships for both self- and interactive contingencies, underscoring the role of the dyadic climate in infant cognitive development.

Again, demonstrating the coordination of gaze, touch, and language, Wojcik et al. (2024) leverage longitudinal recordings from the SAYCam corpus to investigate referent-oriented actions in naturalistic parent-infant communication. They examined 59 h of head-mounted camera video recordings from the viewpoint of a single English-learning infant ($N = 1$) in Australia between ages 6 and 12 months. They coded each caregiver utterance that contained any of nearly 200 early learned words (over 6000 tokens) for 11 variables, including parent and infant gaze, touching, pointing, and reaching to referents. They found that, while referents were generally visible to the infant when words were spoken, there was great variability across word types (e.g., nouns, verbs, adjectives). They also saw that the individual trajectories of referent-oriented cues varied widely across words. This finding suggests that referent-oriented behaviors may not be reliable cues to meaning in the first year of life (perhaps emerging in the second year), and that there may be no universal trajectory of referent cues across different words and word groups. The authors also underscore the value of case studies for generating new hypotheses.

## 3. Coordinating and sharing attention during interaction with others

Coordinated interaction requires coordinated attention. Masek et al. (2024) shed light on the links between infant attention, social contingency, and vocabulary in their examination of $N = 106$ U.S. English- and Spanish-learning infants at 6, 12, and 18 months. Combining experimental, observational, and survey measures, they find that infants' attention to social and non-social stimuli is linked across the first year of life to their social contingency during natural interaction. Further, infant social attention and social contingency were found in path model analyses to predict different aspects of children's later parent-reported vocabulary size. The authors use these findings to argue for early reciprocal development between social and non-social attention, conceptualizing both types of attention as foundational to language development in the first years of life.

Focusing on joint attention, Wildt and Rohlfing (2024) examined how mothers scaffold their infants' gaze following during joint attention episodes (at 6 months), and how this relates to children's later expressive vocabulary development (at 24 months). In a longitudinal sample of $N = 15$ Polish mother-infant dyads, mothers interrupted a typical interaction by directing their gaze away to a different target. Researchers coded mothers' subsequent responses to their infants' behavior, including switching their gaze and monitoring the infant's gaze following. While infants' gaze following was correlated with children's later vocabulary scores, maternal gaze monitoring behaviors were an even stronger predictor of child expressive vocabulary size. These findings emphasize how caregiver's gaze responsiveness might play an important role in scaffolding infants' own gaze following, with implications for their later language development.

Centering digital communication, Myers et al. (2024) examined how infants and toddlers produce and respond to bids for joint visual attention (JVA) in their longitudinal study of video chats between grandparents and grandchildren. While screen-mediated conversation is an increasingly common part of children's upbringing, it brings substantial complications to establishing joint attention, e.g., because manual pointing and gaze direction have to be interpreted indirectly. The authors sought to understand when screen-mediated JVA emerges and how its use is shaped by age and interactional context. $N = 51$ U.S. and Canadian families with young children (4–15 months at time point 1; primarily English speaking) were recorded while making a video call with a grandparent every 2 months for 6 months (3 times maximum). They found that, overall, infant attention to the screen and use of JVA was low, but included both within- and across-screen bids. Higher rates of JVA were associated with greater grandparent sensitivity (Emotional Availability Scales) and with sessions including fewer people. The findings underscore the role of adult sensitivity in eliciting JVA bids across different interactional contexts, and engage broadly with questions regarding children's ability to meaningfully engage with remote family members in a digitally connected world.

## 4. Variation in interactional development

Two sources of variation in interactional development are touched on in this special issue. First, Plate and Iverson (2024) investigate how infants with and without elevated risk for Autism recognize communication breakdowns with caregivers and attempt to repair their intended communication. A sample of $N = 64$ 18-month-old U.S. English-learning infants (49 with elevated likelihood due to having an autistic older sibling [EL], and 15 with typical likelihood for autism [TL]) and a primary caregiver engaged in 15-minutes of free play in their homes. Later, when children were 36-months-old, those with an elevated likelihood of autism were classified as either having Autism (EL-AUT; $n = 9$), a language delay (EL-LD; $n = 15$), or no diagnosis (EL-ND; $n = 25$). Across all four groups, infants initiated communication, experienced breakdowns, and made successful repairs at similar rates. However, there were between-group differences in the types of communication bids infants made and in their repair strategies. Specifically, EL infants bid more for behavior regulation than joint attention, while TL infants preferred the opposite; and EL-AUT infants relied on a smaller repertoire of repair strategies. These findings may inform social communication interventions, especially those focused on leveraging infants' communicative strengths.

Second, Fields-Olivieri et al. (2024) shed light on how a child-level factor—child temperament—influences parent-toddler conversation engagement and children's subsequent expressive language development. In a longitudinal study of $N = 120$ economically strained, English-speaking families living in rural and semi-rural northeastern U.S. areas, they evaluated each toddler for challenging temperament (high negative affect and low effortful control) at 18 and 30 months, the contingency of parent-child conversation during free play or book reading at 18 and 30 months, and the complexity and diversity of children's speech samples at 18, 24, and 36 months. A path model indicated that children with more challenging temperament experienced less conversational engagement at both 18 and 30 months, which in turn predicted lower expressive language skills at 24 and 36 months. These findings highlight the importance of investigating child-level factors in addition to family-level factors (such as demographics) when trying to understand why families vary in their conversational dynamics.

The twelve articles in this special issue thus collectively yield key insights into how children's language development is impacted by the interactions they engage in from early infancy, and how infants' developing sense of self is shaped by their interactional roles with respect to people, places, and activities in their everyday life.

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